MILITARY SPECIFICATION

DUMMY CARTRIDGE, 5.56MM. XM199

1. SCOPE

1.1 This specification covers a dummy cartridge used for training personnel and for inspecting and testing the mechanism of 5.56 small arms.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-I-45208 — Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-644 — Visual Inspection Standards and Inspection Procedures for Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.

DRAWINGS

MUNITIONS COMMAND

C10534021 — Dummy Cartridge, 5.56MM, XM199.

F10523635 — Packing and Marking, Cartridges, 5.56MM; Cartons; Box, Ammunition, M2A1; Box, Wirebound.

IEL-10534021 — Inspection Equipment List for Dummy Cartridge, 5.56MM, XM199.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **General.** The cartridge shall comply with Drawing C10534021, referenced specifications and the following:

3.2 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 **Bullet extraction.** The force required to extract the bullet from the cartridge case shall be not less than 175 pounds.

3.4 **Residual stress.** The cartridge case shall not split when subjected to a 1% mercurous nitrate solution for 15 minutes.

3.5 **Workmanship.** The requirements for workmanship are as specified on the appli-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.

cable drawings, referenced specifications and the following:

3.5.1 *Metal defects.* The cartridge shall be free of splits.

3.5.2 *Foreign matter.* The cartridge shall be free of corrosion, dirt and oil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 for definition of quality assurance terms.

4.1.2 *Inspection.* Inspection shall be in accordance with Specification MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the initiation of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 100 cartridges. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing must be the same as will be used for regular production and shall be obtained from the same source of supply. The supplier shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with Standard MIL–STD–105.

4.3.1.2 *Lot identification.* Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 *Examination.* Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, using applicable sampling plans and acceptance criteria of Standard MIL–STD–105. The acceptable quality level (AQL) for the major class shall be 1.00 percent and the AQL for the minor class shall be 2.50 percent.

4.3.2.1 *Classification of defects.* The classification of defects shall be as follows:

	Defect	Method of inspection
Major		
101.	Cartridge profile failure (requiring more than 10 pounds dead weight to insert in profile and alignment gage).	Gage
102.	Total length, max.	Gage
103.	Split case	Visual
104.	Loose bullet	Manual
Minor		
201.	Absence of corrugation	Visual
202.	Corrosion, dirt or oil	Visual
203.	Split bullet jacket	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.

4.3.3 *Tests.* The following tests shall be conducted in accordance with the methods and procedures specified in 4.4, except as provided in Specification MIL–I–45208.

4.3.3.1 *Bullet extraction.* A sample of fifty cartridges shall be tested for bullet extraction. Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

4.3.3.2 *Residual stress (Mercurous nitrate).* A sample of twenty cartridges shall be subjected to the mercurous nitrate test. Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

4.3.4 *Packaging, packing, and marking inspection.* During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that no primer is present in the primer pocket. The presence of a primer shall be classed as a critical defect and all cartridges containing a primer shall be rejected. Inspection for packaging, packing and marking shall be in accordance with Standard MIL–STD–644 as applicable to the drawing or as required by the contract.

4.3.5 *Inspection equipment.* The examination and tests shall be made using the equipment listed on IEL–10534021, except as provided in Specification MIL–I–45208.

4.4 **Test methods and procedures.**

4.4.1 *Bullet extraction.* The test shall be conducted in accordance with Appendix B. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 *Residual stress (Mercurous nitrate).* The test shall be conducted in accordance with Appendix A.

5. PREPARATION FOR DELIVERY

5.1 **Packing.** *Level A (Worldwide shipment).* The cartridges shall be packed in accordance with Drawing F10523635, or as required by contract.

5.2 **Marking and labeling.** Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 **Ordering data.** Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

Custodian:
Army—MU

Preparing activity:
Army—MU

Project No. 1305–A288

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.

APPENDIX A

MERCUROUS NITRATE TEST PROCEDURE

1. EQUIPMENT

1.1 Equipment shown on drawings listed in the Mercurous Nitrate section of the Inspection Equipment List shall be used.

2. MANDATORY SAFETY REQUIREMENTS

2.1 Food shall not be stored or eaten in vicinity in which these tests are conducted.

2.2 Acid resistant apron and gloves or the equivalent shall be worn by each test technician.

2.3 Face shields shall be worn at all times during the pouring or mixing of acids and water. Safety glasses shall be worn during other phases of this test.

2.4 Asbestos or heat insulating gloves shall be worn during the heat volatilization phase of the test to facilitate handling.

2.5 During the entire period of volatilization the oven door shall be closed.

2.6 Extreme care shall be exercised in the mixing of acid with water; this shall be accomplished by pouring the acid into the water.

2.7 Test shall be conducted under a canopy or hood having a forced draft ventilation system to remove the noxious fumes. The disposition of tested components shall be governed by local regulations.

3. TEST PROCEDURE

3.1 Preparation of solution.

3.1.1 Nitric-Acid Solution: Four hundred (400) milliliters of nitric acid (of specific gravity 1.42) are dissolved in five hundred (500) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter. (The resultant specific gravity will be 1.25).

3.1.2 Mercurous-Nitrate Solution: Ten (10) grams of mercurous nitrate and ten (10) milliliters of nitric acid (of specific gravity 1.42) are dissolved in four hundred (400) milliliters of distilled water at room temperature. To this solution, distilled water is added to bring the volume of the resulting solution to one liter.

3.2 Test of cartridges.

3.2.1 The cartridges are positioned in both nitric acid and mercurous nitrate solutions in a vertical position, with the head of the case down. The depth of each solution is adjusted until it completely covers the mouth of the case.

3.2.2 The number of cartridges prescribed for the Mercurous Nitrate test is submerged in the nitric acid solution. After thirty (30) seconds the cartridges are withdrawn, rinsed in water and the excess water removed. The cartridges are then submerged for fifteen (15) minutes in the mercurous nitrate solution. Upon removal, the mercury on the surface of the case is volatilized by the application of heat and the surface then examined under magnification (10 to 15 diameters) for splits and cracks. All splits and cracks found are reported.

Note. A split is defined as a separation of the metal entirely through the wall of the case. A crack is a surface condition and represents a separation of the metal partially through the wall. Cracks are not considered to be splits. A suggested method for determining cracks and splits follows: Splits in the cartridge case, after the mercury cracking test, are

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.

defined by extracting the bullet, filling the case with water (water temperature 70° F. to 100° F.) to the mouth until a convex meniscus condition exists and placing the thumb of the hand over the mouth of the case. If the case is split this exerts sufficient pressure to force the water through the opening.

4. RECORDING DATA

4.1 Results of the test shall be recorded, including all splits and cracks and their location in the case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.

APPENDIX B

BULLET EXTRACTION TEST PROCEDURE

1. EQUIPMENT

1.1 Equipment shown on drawings listed in the Bullet Extraction section of the Inspection Equipment List shall be used.

2. TEST PROCEDURE

2.1 Preparation for test.

2.1.1 The testing machine should be calibrated either weekly or prior to each occasion of use, whichever is less frequent. Calibration points shall include at least 30 pounds, 60 pounds, 100 pounds, 150 pounds and 250 pounds; if calibration errors exceed three pounds at scale readings below 200 pounds or six pounds at scale readings above 200 pounds, then the necessary corrective action shall be taken to reduce the errors below these limits before the bullet-extraction test is conducted. The method of calibration shall be by calibrated proof rings or calibrated spring balances which shall yield values within $\pm 1\%$ of true values over calibrating range.

2.1.2 The number of cartridges as prescribed for this test are placed in a recessed holding block which shall be placed at a point convenient to the technician.

2.2 Conducting the test.

2.2.1 The cartridge shall be inserted into the case holding block on the pulling head and aligned with the jaws. The jaws shall then be secured to the bullet just above the neck of the case, and the load applied.

2.2.2 When the bullet has been extracted from the case, the machine shall be stopped, and the force required to extract the bullet from the case, indicated by the tensile indicator, recorded.

2.2.3 The case and bullet shall then be removed from the holding block and jaws.

3. RECORDING OF DATA.

3.1 Force required to extract each bullet.

3.2 Grand average of the individual values.

3.3 Minimum value.

3.4 Maximum value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:51Z
Check the source to verify that this is the current version before use.